

Problem

Express the following functions in cosine form:

(a) $10 \sin(\omega t + 30^\circ)$

(b) $-9 \sin(8t)$

(c) $-20 \sin(\omega t + 45^\circ)$

Step-by-step solution

Step 1 of 3

Convert the following functions in cosine form:

(a)

Consider the function is $10 \sin(\omega t + 30^\circ)$.

$$\begin{aligned} f &= 10 \sin(\omega t + 30^\circ) \\ &= 10 \cos(\omega t + 30^\circ - 90^\circ) \\ &= 10 \cos(\omega t - 60^\circ) \end{aligned}$$

Hence the cosine form is $10 \cos(\omega t - 60^\circ)$.

Step 2 of 3

(b)

Consider the function is $-9 \sin(8t)$.

$$\begin{aligned} f &= -9 \sin(8t) \\ &= 9 \cos(8t + 90^\circ) \end{aligned}$$

Hence the cosine form is $9 \cos(8t + 90^\circ)$

Step 3 of 3

(c)

Consider the function is $-20 \sin(\omega t + 45^\circ)$

$$\begin{aligned} f &= -20 \sin(\omega t + 45^\circ) \\ &= 20 \cos(\omega t + 45^\circ + 90^\circ) \\ &= 20 \cos(\omega t + 135^\circ) \end{aligned}$$

Hence the cosine form is $20 \cos(\omega t + 135^\circ)$.